CLOUD SECURITY SPECIALIST
NAME:KEMUNTO FAITH OGWOKA
CS NO:ADC-CSS02-25071
GMAIL:faithogwoka@gmail.com
LAB:AZURE FIREWALL

1.1 INTRODUCTION

I started the lab with a required lab set up.I clicked on ctrl alt and delete to bring up the login page,signed in as admin and then downloaded the lab files.

1.2 LAB 03:AZURE FIREWALL

EXERCISE 1:DEPLOY AND TEST AN AZURE FIREWALL

In this exercise I used a template to deploy the lab environment,deployed an Azure firewall,created a default route,configured an application rule,configured a network rule,configured DNS servers and tested the firewall.

TASK 1

In this task,I created a virtual machine by using an ARM template.

Step 1

I signed into the Azure portal with the credentials provided by the lab owner.

Step 2

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page,I typed Deploy a custom template and pressed the Enter key.

Step 3

On the Custom deployment blade, I clicked on the Build your own template in the editor option

Step 4

On the Edit template blade, I clicked the Load file, located the \Allfiles\Labs\08\template.json file and clicked Open.

Step 5

I clicked save on the edit template blade

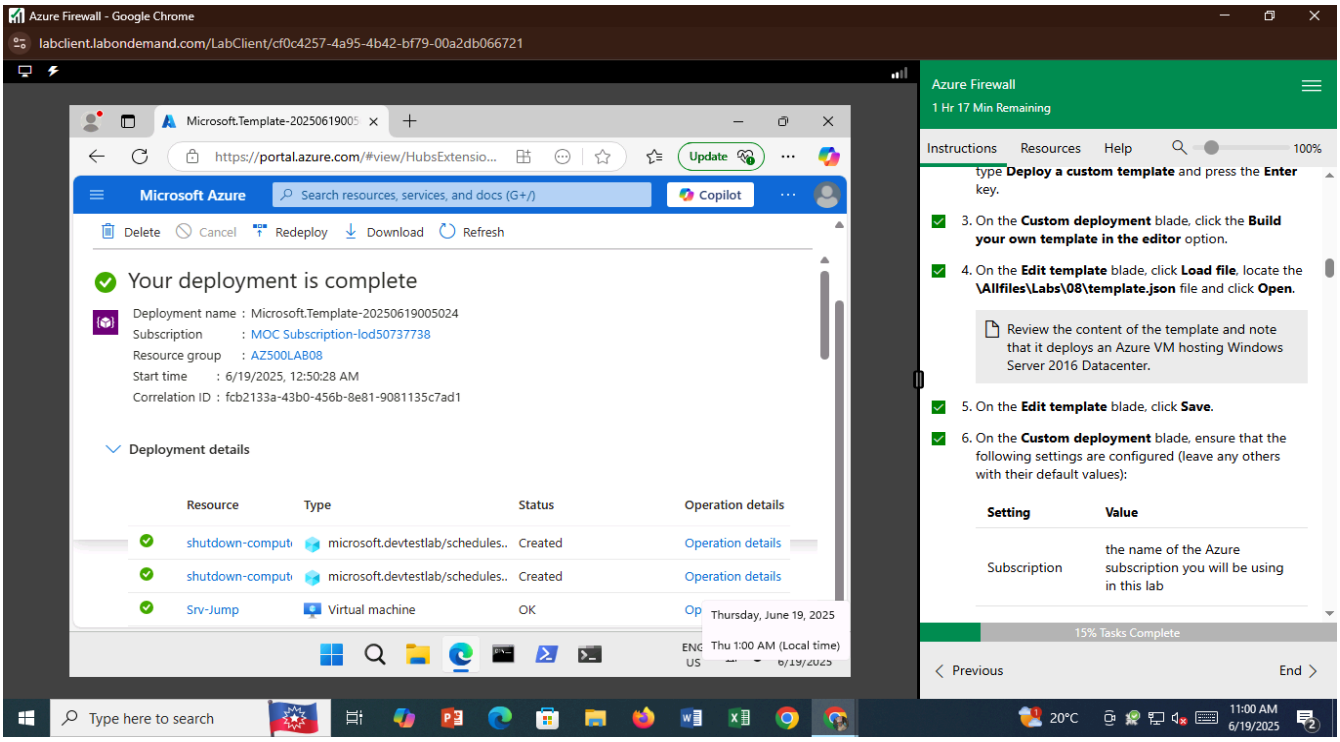
Step 6

On the Custom deployment blade, I ensured that the settings provided for;subscription,resource group,location and admin password were configured.

Step 7

I clicked on Review + create,waited for the deployment to finish and then clicked on the Create button.

Here is my screenshot



TASK 2

In this task, I deployed the Azure firewall into the virtual network

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Firewalls and pressed the Enter key.

Step 2

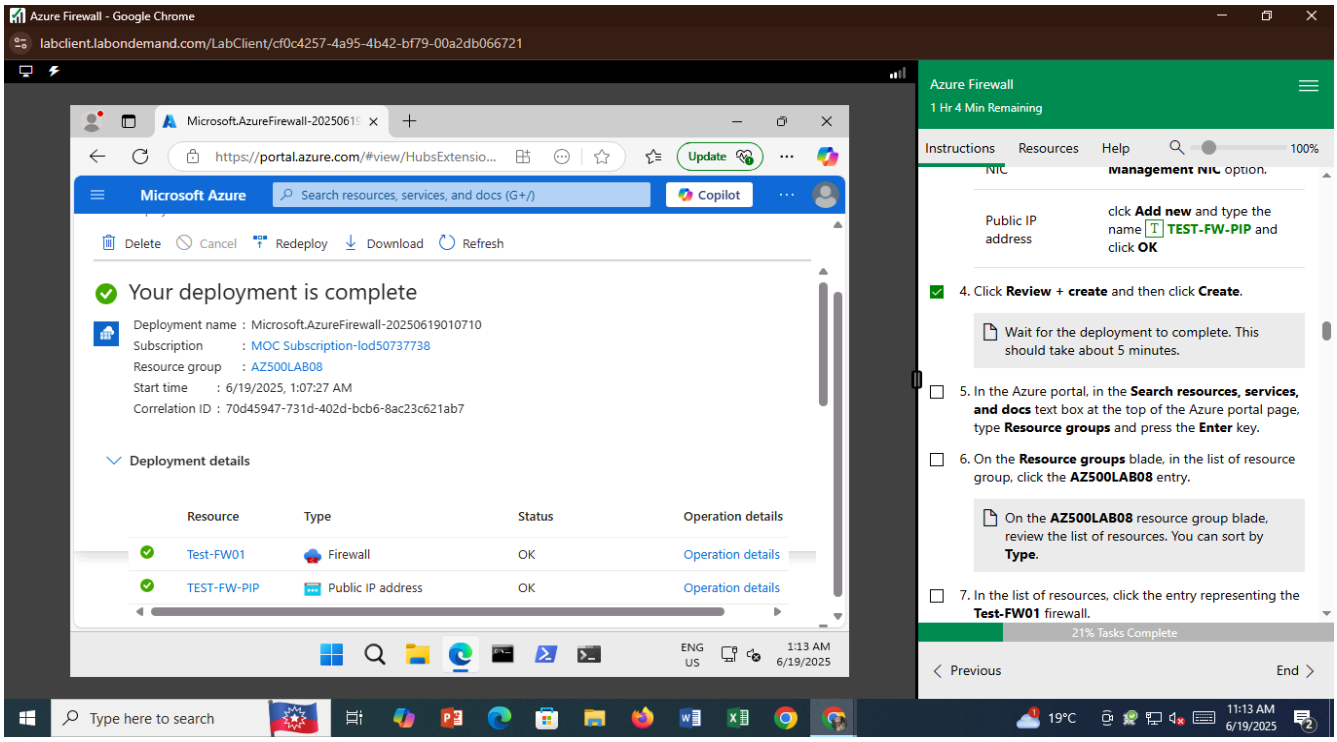
On the Firewalls blade,I clicked + Create

Step 3

On the Basics tab of the Create a firewall blade, I specified the settings provided

Step 4

I clicked the Review + create button, then Create and waited for the deployment to complete.Below is my screenshot



Step 5

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Resource groups and pressed the Enter key

Step 6

On the Resource groups blade, in the list of resource group,I clicked the AZ500LAB08 entry

Step 7

In the list of resources,I clicked the entry representing the Test-FW01 firewall.

Step 8

On the Test-FW01 blade, I identified the Private IP address that was assigned to the firewall.

TASK 3

In this task, I created a default route for the Workload-SN subnet. This route would configure outbound traffic through the firewall.

Step 1

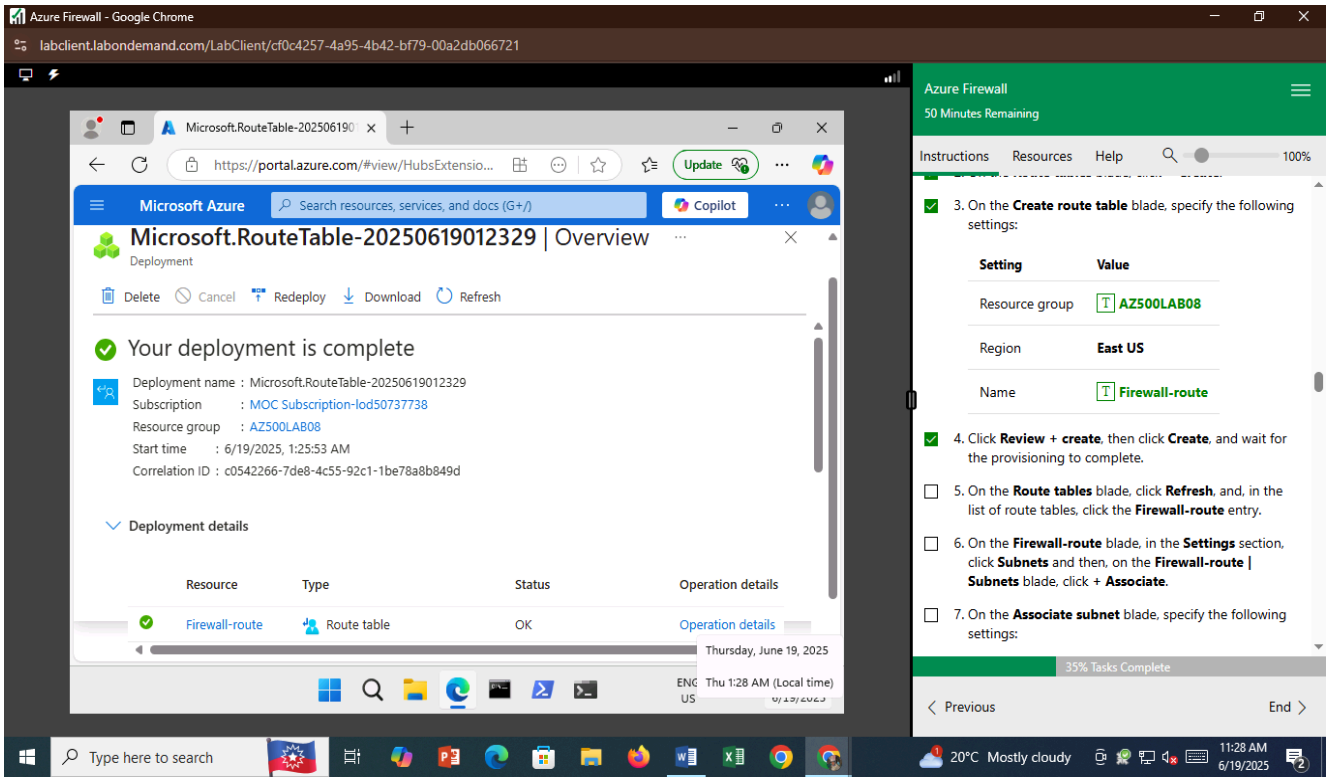
In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Route tables and pressed the Enter key.

Step 2 and 3

On the Route tables blade,I clicked on the + Create button and specified the settings provided.

Step 4

I clicked Review + create, then clicked Create, and waited for the provisioning to complete.Below is my screenshot



Step 5

On the Route tables blade, I clicked on Refresh, and, in the list of route tables, clicked the Firewall-route entry.

Step 6

On the Firewall-route blade, in the Settings section, I clicked on Subnets and then, on the Firewall-route | Subnets blade, click + Associate.

Step 7

On the Associate subnet blade, I specified the settings provided for virtual network and subnet.

Step 8

I clicked OK to associate the firewall to the virtual network subnet.

Step 9

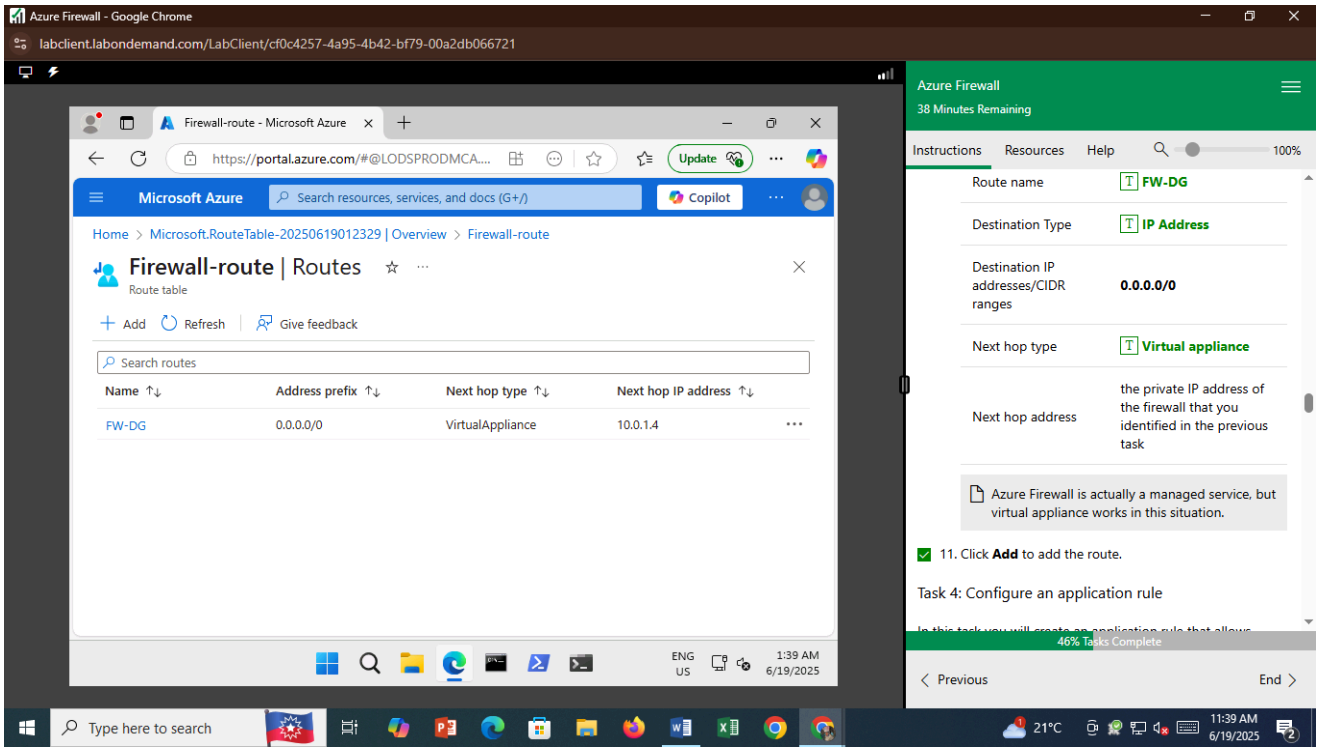
Back on the Firewall-route blade, in the Settings section,I clicked Routes and then + Add.

Step 10

On the Add route blade, I specified the settings provided

Step 11

I clicked the Add button to add the route.Below is my screenshot



TASK 4

In this task, I created an application rule that allows outbound access to www.bing.com

Step 1

In the Azure portal, I navigated back to the Test-FW01 firewall

Step 2

On the Test-FW01 blade, in the Settings section, I clicked on Rules (classic)

Step 3

On the Test-FW01 | Rules (classic) blade, I clicked the Application rule collection tab, and then clicked the + Add application rule collection.

Step 4

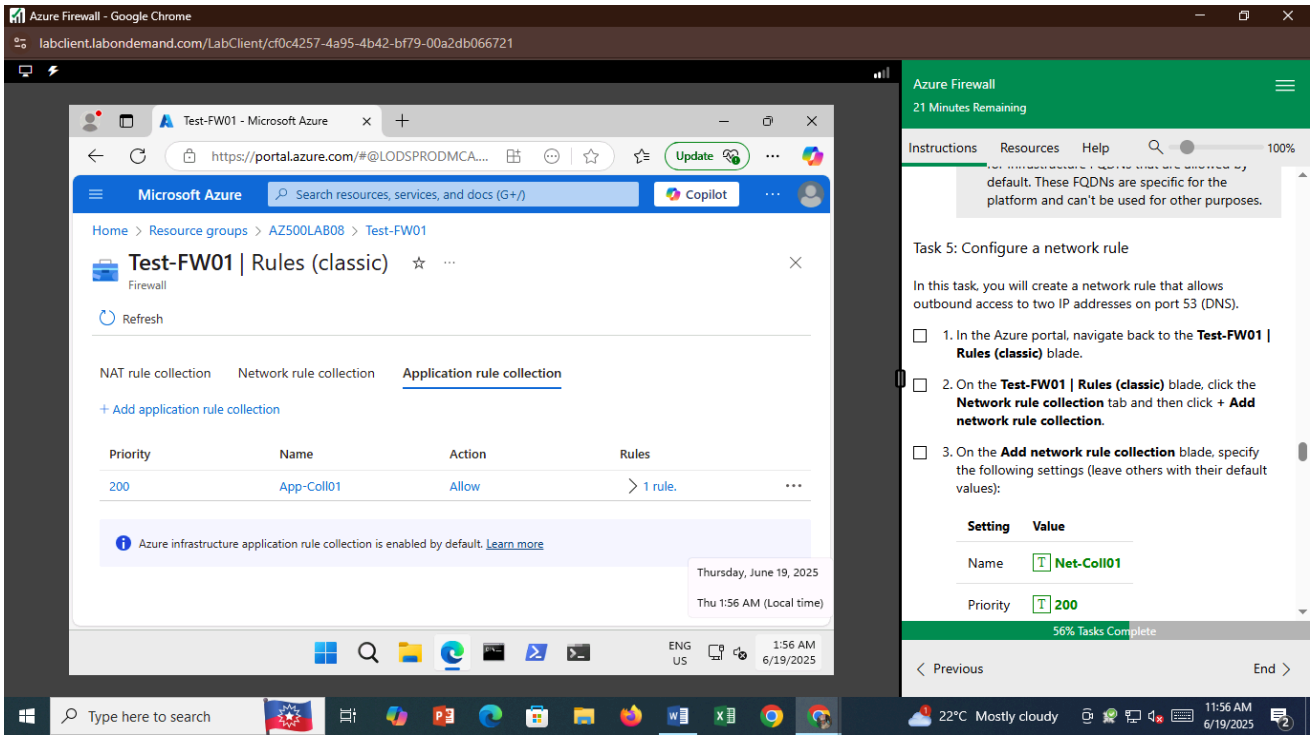
On the Add application rule collection blade, I specified the settings provided for name,priority, action and left the rest as default.

Step 5

On the Add application rule collection blade, I created a new entry in the Target FQDNs section with the settings provided.

Step 6

I clicked the Add button to add the Target FQDNs-based application rule.Below is my screenshot



TASK 5

In this task,I created a network rule that allows outbound access to two IP addresses on port 53 (DNS).

Step 1

In the Azure portal, I navigated back to the Test-FW01 | Rules (classic) blade

Step 2

On the Test-FW01 | Rules (classic) blade, I clicked the Network rule collection tab and then clicked + Add network rule collection.

Step 3

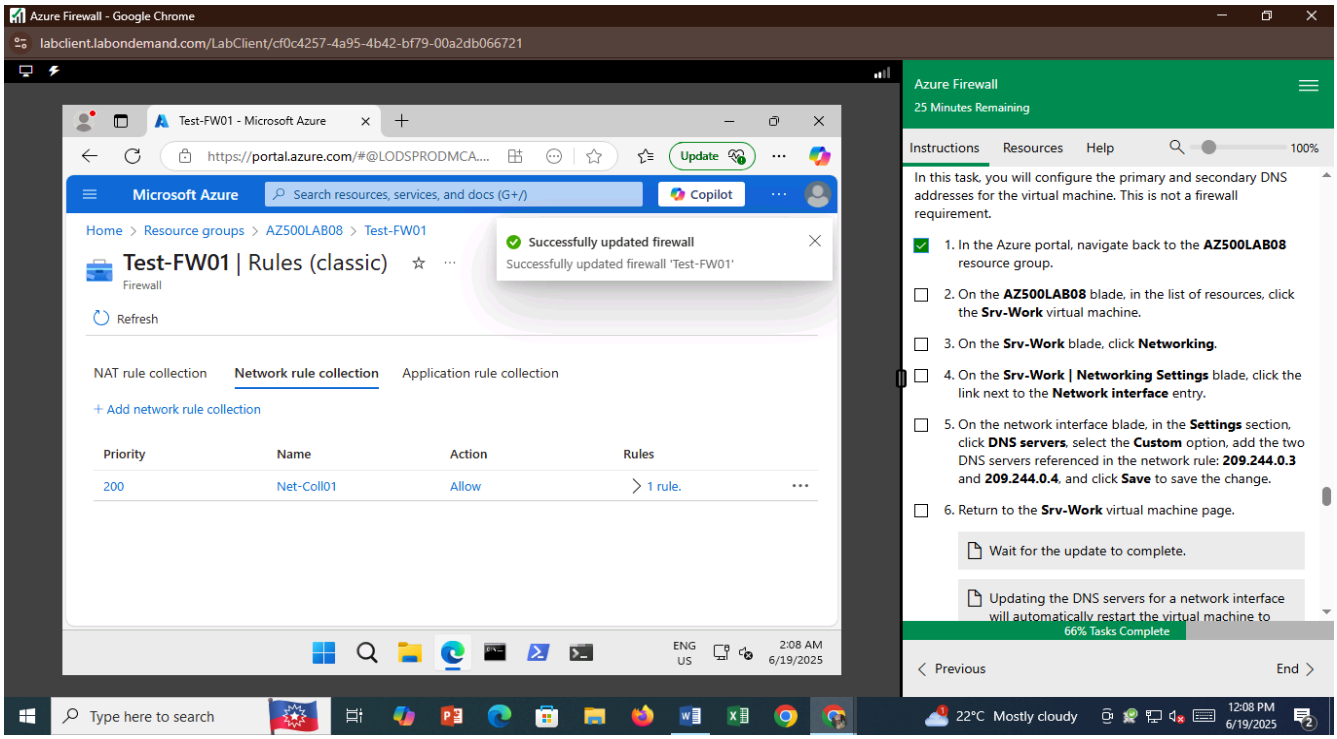
On the Add network rule collection blade, I specified the settings provided for name,priority and action.

Step 4

On the Add network rule collection blade, I created a new entry in the IP Addresses section with the settings provided

Step 5

I clicked the Add button to add the network rule.Below is my screenshot



TASK 6

In this task, I configured the primary and secondary DNS addresses for the virtual machine.

Step 1

In the Azure portal,I navigated back to the AZ500LAB08 resource group

Step 2

I clicked the Srv-Work virtual machine on the AZ500LAB08 blade, in the list of resources.

Step 3

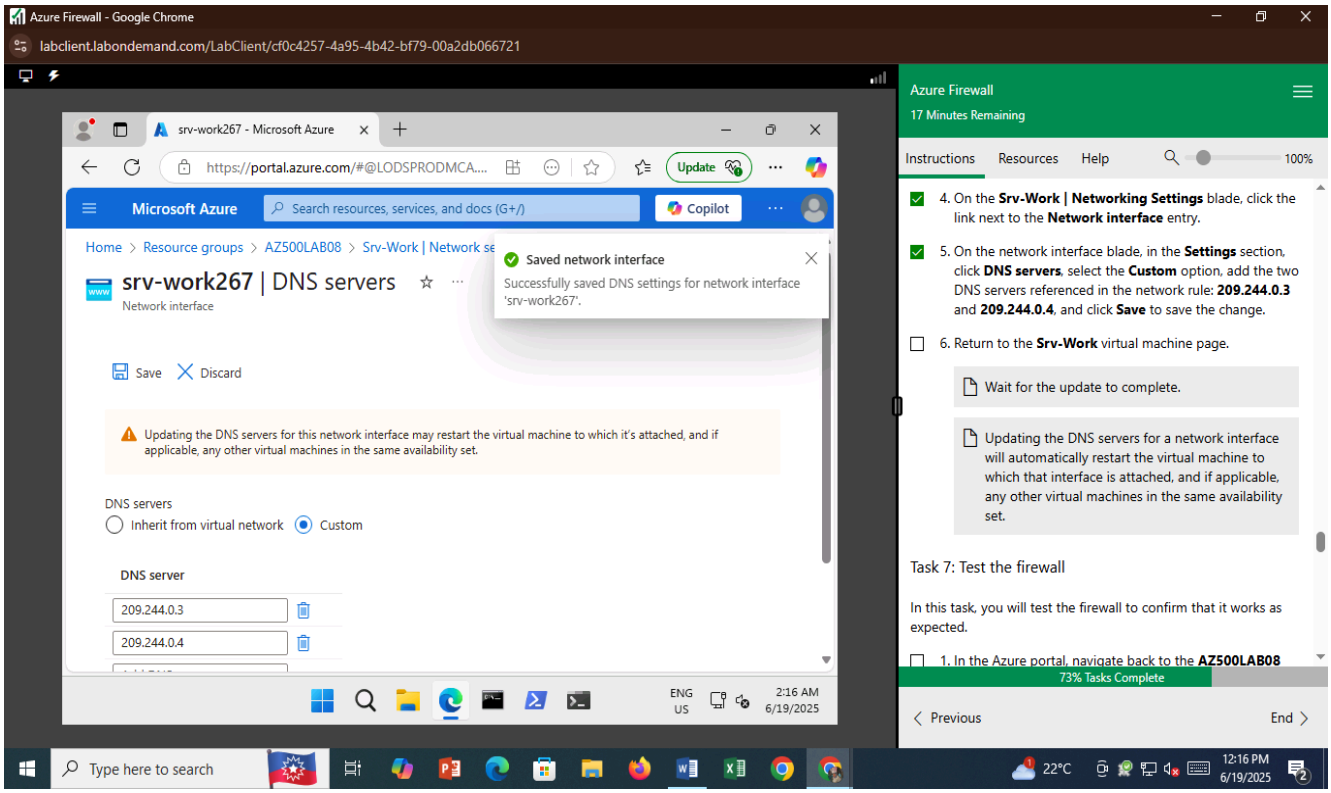
On the Srv-Work blade, I clicked on Networking.

Step 4

On the Srv-Work | Networking Settings blade, I clicked the link next to the Network interface entry

Step 5

On the network interface blade, in the Settings section, I clicked DNS servers, selected the Custom option, added the two DNS servers referenced in the network rule: 209.244.0.3 and 209.244.0.4, and clicked Save to save the change.Below is my screenshot



Step 6

I returned to the Srv-Work virtual machine page.

TASK 7

In this task, I tested the firewall to confirm that it works as expected

Step 1

In the Azure portal,I navigated back to the AZ500LAB08 resource group.

Step 2

On the AZ500LAB08 blade, in the list of resources, I clicked the Srv-Jump virtual machine

Step 3

On the Srv-Jump blade, I clicked connect and, in the drop down menu, clicked RDP

Step 4

I clicked Download RDP File and used it to connect to the Srv-Jump Azure VM via Remote Desktop. When prompted to authenticate, provided the credentials

Step 5

Within the Remote Desktop session to Srv-Jump, I right-clicked Start, in the right-click menu, clicked Run, and, from the Run dialog box, ran the command provided to connect to Srv-Work.

Step 6

When prompted to authenticate, I provided the credentials to sign in with.

Step 7

Within the Remote Desktop session to Srv-Work, in Server Manager, I clicked on the Local Server and IE Enhanced Security Configuration.

Step 8

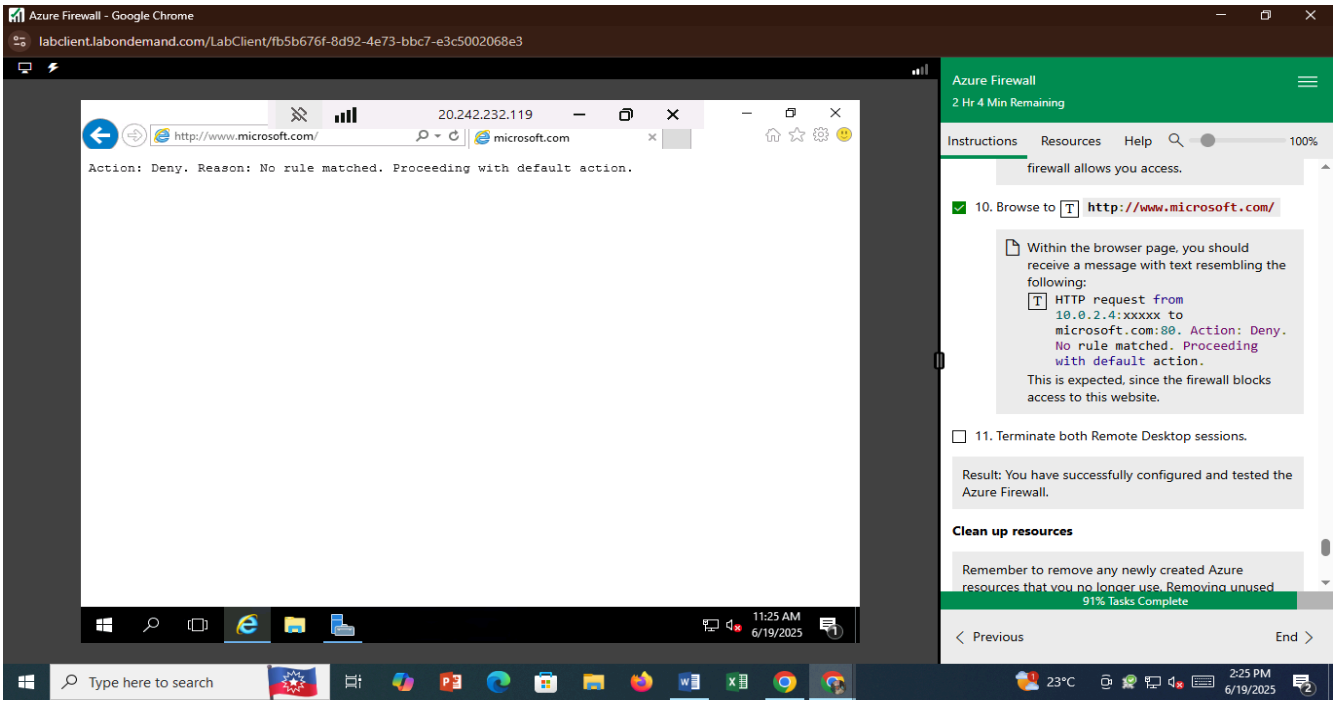
In the Internet Explorer Enhanced Security Configuration dialog box,I set both administrators and users options to Off and clicked OK.

Step 9

Within the Remote Desktop session to Srv-Work,I started Internet Explorer and browsed to https://www.bing.com.

Step 10

I browsed to http://www.microsoft.com/ and received the message,Action: Deny. No rule matched. Proceeding with default action.I learned that this is expected since the firewall blocks access to this website.Here is a screenshot



Step 11

I terminated both Remote Desktop sessions,cleaned up my resources and completed the lab.